

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think> The assistant is thinking about the fact that B cells play a variety of roles in the immune response , including the production of antibodies and the activation of other immune cells . The cluster of differentiation 21 (CD 21) is a protein that is found on the surface of B cells and is involved in the process of antigen presentation . It is also a receptor for the complement protein C 3 d , which is involved in the activation of B cells . By genetically modifying B cells so that they do not express CD 21 , the scientists are likely to be affecting the ability of B cells to activate and respond to a variety of antigens . This could have a variety of effects , including a reduced ability to produce antibodies and a reduced ability to activate other immune cells . The pathogenesis of a variety of organisms could be affected by this genetic modification , including viruses that are able to infect B cells . </think><answer> B . Measles virus </answer>

